

Student Placement Readiness Management System using Salesforce

ABSTRACT

This document describes the design and implementation of a **Student Placement Readiness Management System** developed using the Salesforce platform. The primary objective of this project is to automate the evaluation of student placement preparedness based on **technical skills, professional certifications, and academic performance (CGPA)**.

As educational institutions scale, manual tracking of placement readiness becomes inefficient and error-prone. This Salesforce-based solution leverages **custom objects, declarative automation using Salesforce Flows, validation rules, reports, and dashboards** to ensure accurate assessment and real-time visibility of student readiness levels. The system classifies students into **Ready, Almost Ready, and Not Ready** categories, helping placement coordinators make informed decisions efficiently without relying on complex programming.

OBJECTIVE

The core objective of this project is to design and implement a centralized Salesforce system to manage and evaluate student placement readiness.

Specific goals include:

- Automating placement readiness score calculation.
- Eliminating manual evaluation errors.
- Ensuring consistent and structured student data.
- Providing real-time readiness insights to placement coordinators.
- Improving coordination between students, academic performance, and placement activities.
- Visualizing readiness data using Salesforce reports and dashboards.

TECHNOLOGY DESCRIPTION

Salesforce Platform

Salesforce is a cloud-based CRM platform that enables rapid application development using both declarative (click-based) and programmatic tools. In this project, Salesforce is used as a centralized platform to manage student data, automate readiness calculations, and visualize placement readiness metrics.

The system primarily uses **Flows instead of Apex**, aligning with Salesforce's *Clicks Not Code* philosophy.

Custom Objects

The following custom objects were created to support the placement readiness process:

Student__c

Stores student personal and academic details.

- Student Name
- Student ID
- Branch
- Semester
- CGPA
- Placement Status

Skill__c

Stores technical skills associated with students.

- Skill Name
- Proficiency Level (Beginner / Intermediate / Advanced)
- Student (Lookup)

Certification__c

Stores professional certifications completed by students.

- Certification Name
- Platform

- Completion Date
- Student (Lookup)

Placement_Readiness__c

Stores aggregated readiness information.

- Skill Score
- Certification Score
- CGPA Score
- Total Readiness Score
- Readiness Level
- Student (Lookup)

TABS AND LIGHTNING APP

Tabs

Custom object tabs were created for:

- Students
- Skills
- Certifications
- Placement Readiness

These tabs allow users to easily create, view, and manage records.

Custom Lightning App

A Lightning App named **Student Placement Dashboard** was created to bring together:

- Student Records
- Skills
- Certifications
- Placement Readiness
- Reports
- Dashboards

This app acts as a centralized workspace for placement coordinators.

VALIDATION RULES

Validation rules were implemented to maintain data integrity:

- Prevent saving Placement Readiness without a linked Student.
- Ensure CGPA values remain within valid academic limits.
- Restrict invalid proficiency selections.

AUTOMATION USING SALESFORCE FLOWS

1. Skill Score Calculation Flow

- Triggered when a Skill record is created or updated.
- Assigns scores based on proficiency level:
 - Beginner → 5
 - Intermediate → 10
 - Advanced → 15
- Updates the corresponding student's Placement Readiness Skill Score.

2. Certification Score Calculation Flow

- Triggered when a Certification record is created.
- Counts the total certifications per student.
- Assigns scores based on count:
 - 1 Certification → 10
 - 2 Certifications → 20
 - 3 or More → 30
- Updates Certification Score in Placement Readiness.

3. Placement Readiness Evaluation Flow

- Automatically calculates Total Readiness Score using:
- $\text{Total Score} = \text{Skill Score} + \text{Certification Score} + \text{CGPA Score}$
- Assigns Readiness Level:
 - $\geq 80 \rightarrow \text{Ready}$
 - $50-79 \rightarrow \text{Almost Ready}$
 - $< 50 \rightarrow \text{Not Ready}$
- Updates Student Placement Status accordingly.

REPORTS AND DASHBOARDS

Reports Created:

- Placement Readiness Score Report
- Ready Students Report
- Branch-wise Placement Readiness
- Student Skill Report
- Student Certification Report

Dashboard:

A dashboard titled **Student Placement Dashboard** displays:

- Readiness Level Distribution (Pie Chart)
- Branch-wise Ready Students (Bar Chart)
- Top Placement-Ready Students (Table)
- Certification Distribution

DETAILED PROJECT EXECUTION

1. Created Salesforce Developer Org.
2. Designed custom objects and relationships.
3. Created tabs for all custom objects.
4. Built Lightning App for placement management.
5. Implemented flows for skill, certification, and readiness calculation.
6. Created reports and dashboards for visualization.
7. Tested the system using sample student data.

PROJECT DEMONSTRATION – REAL SCENARIO

1. A student record is created.
2. Skills are added with proficiency levels.
3. Certifications are added.
4. Salesforce flows automatically calculate scores.
5. Placement readiness level updates in real time.
6. Reports and dashboards reflect the updated readiness instantly.

This demonstrates a fully automated placement readiness evaluation system.

SCREENSHOTS

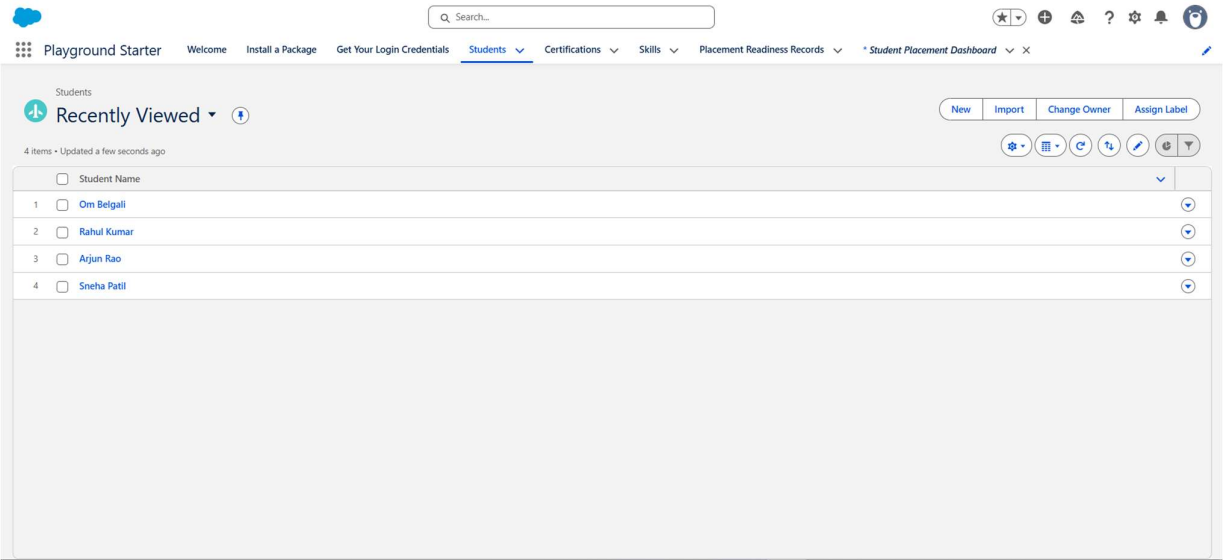


Fig: Student Custom Tab

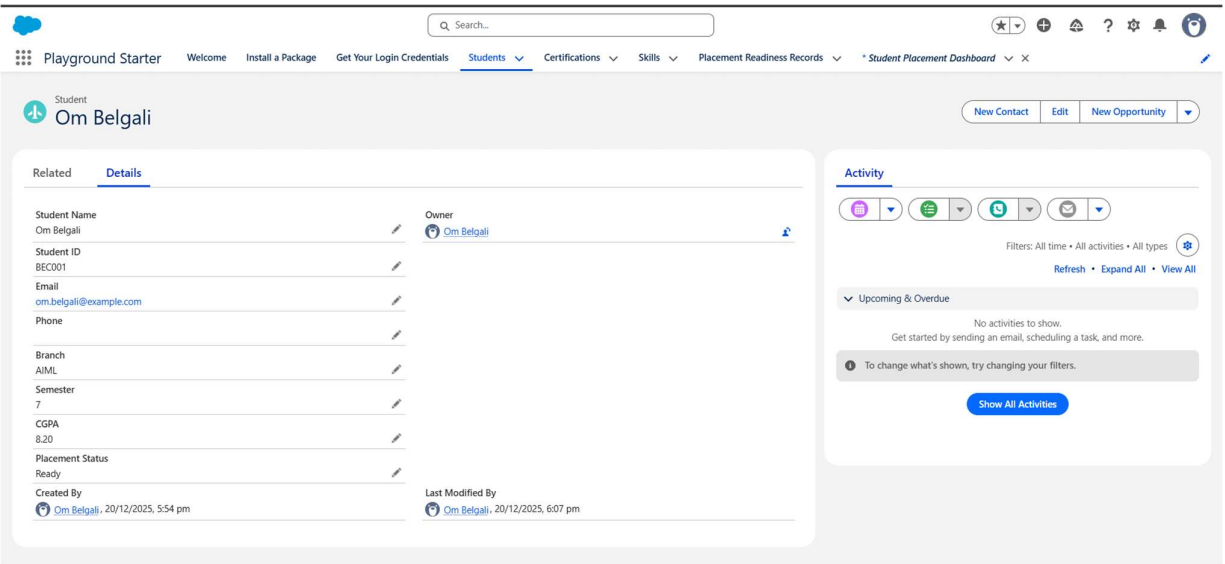


Fig: Student Record Creation Page

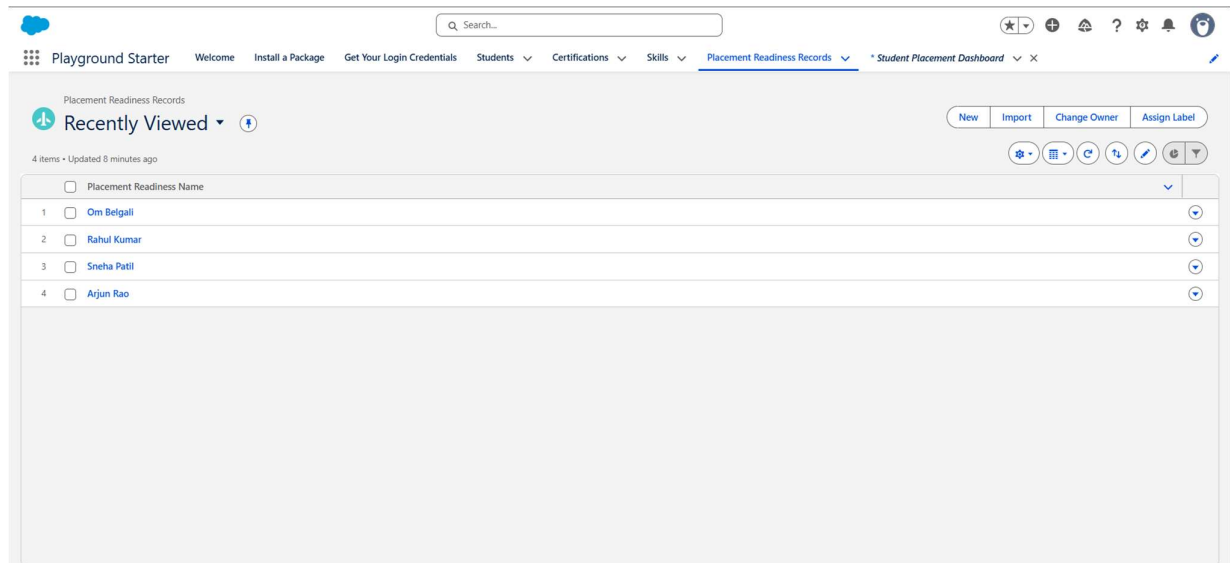


Fig: Placement Readiness Custom Tab

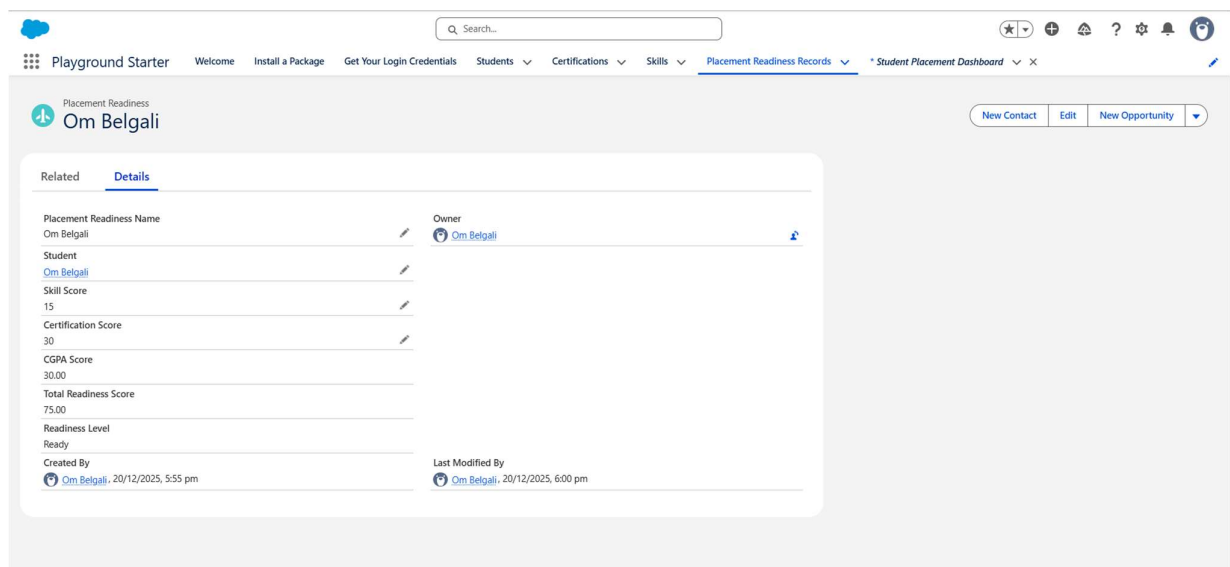


Fig: Placement Readiness Record Creation

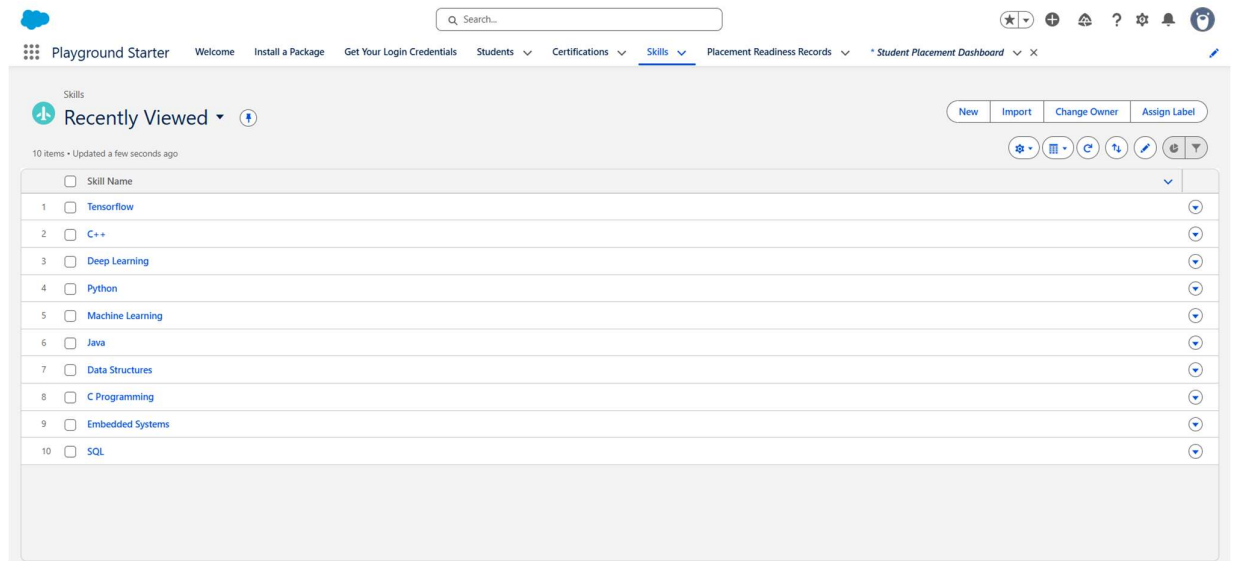


Fig: Skills Custom Tabs

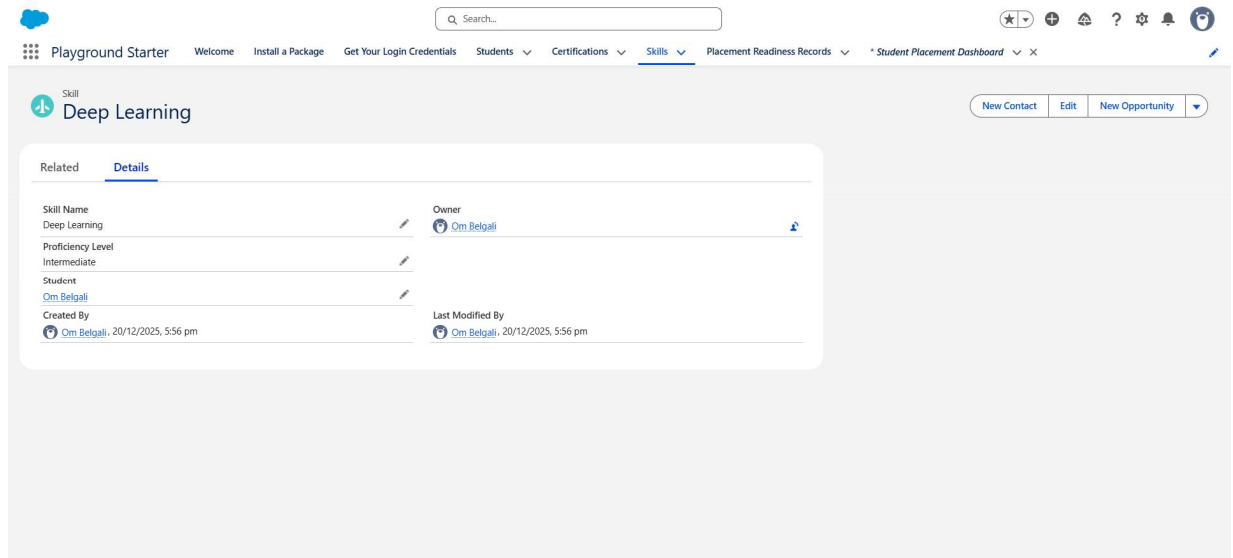


Fig: Skill Record Creation for Student

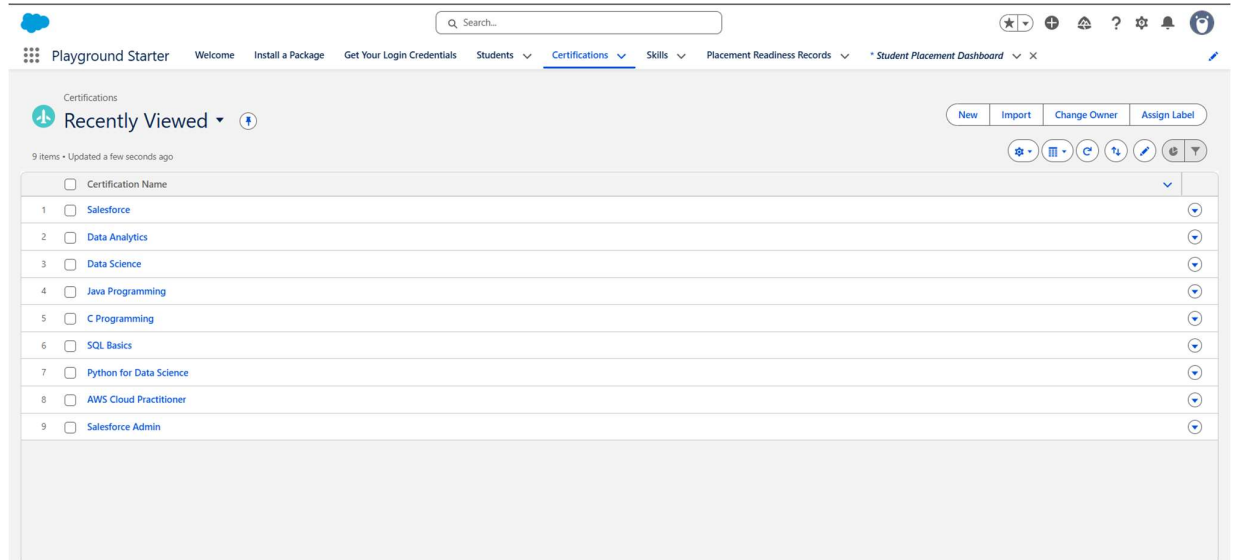


Fig: Certification Custom Tab

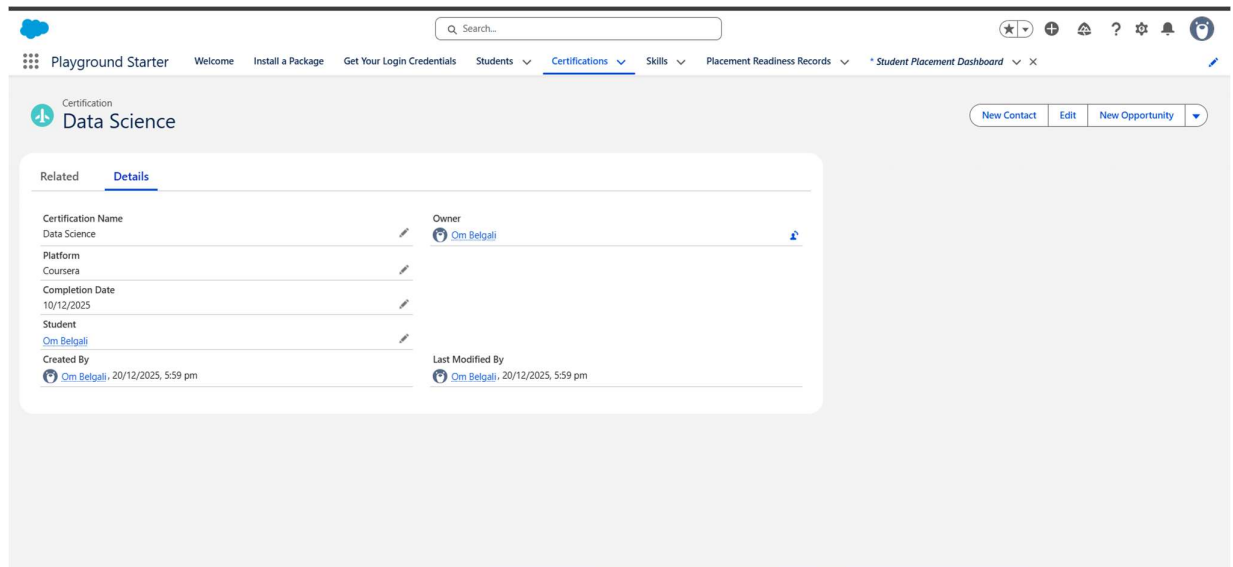


Fig: Certification Record Creation for Student

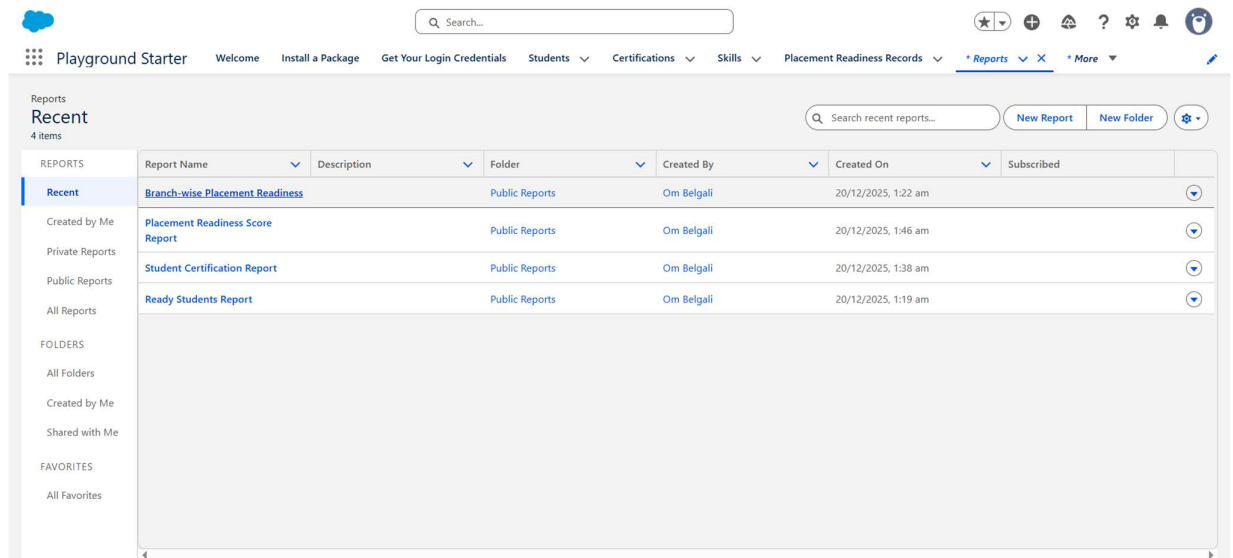


Fig: Reports Tab

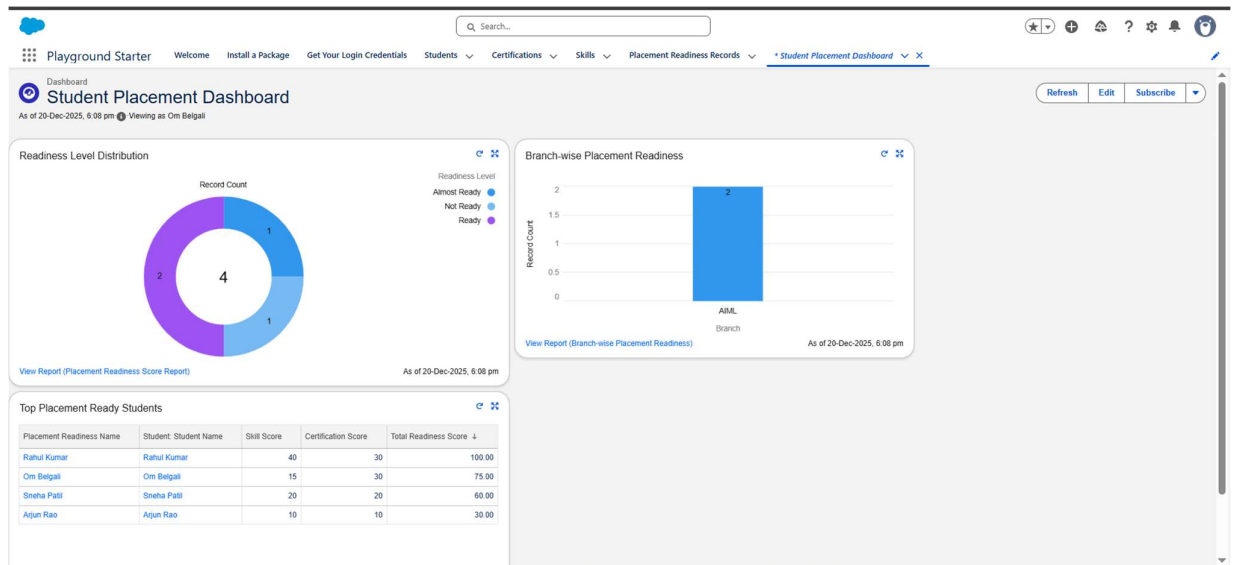


Fig: Student Placement Dashboard Overview

FLOWS

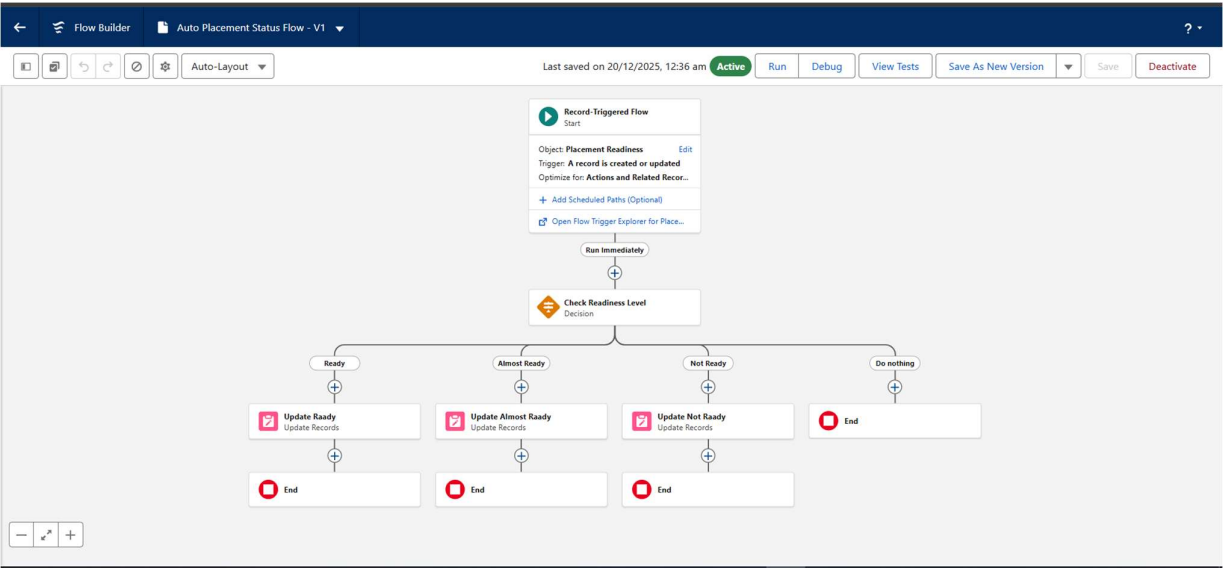


Fig: Automatically Calculated Total Readiness Level

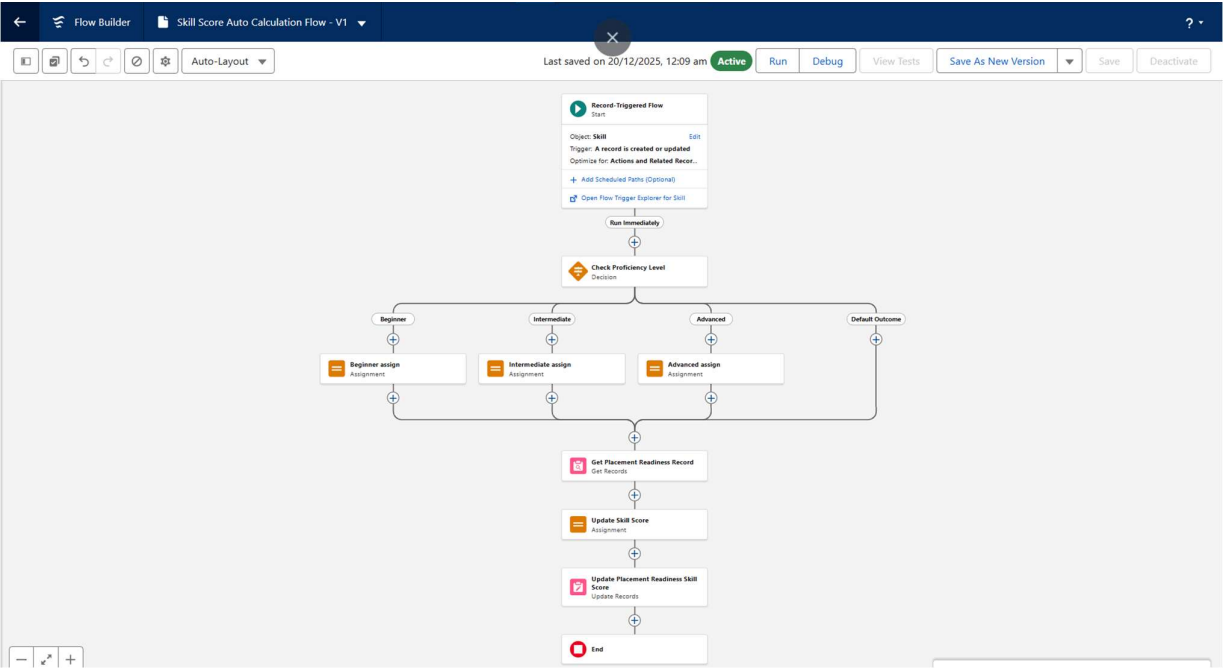


Fig: Placement Readiness Record with Updated Skill Score

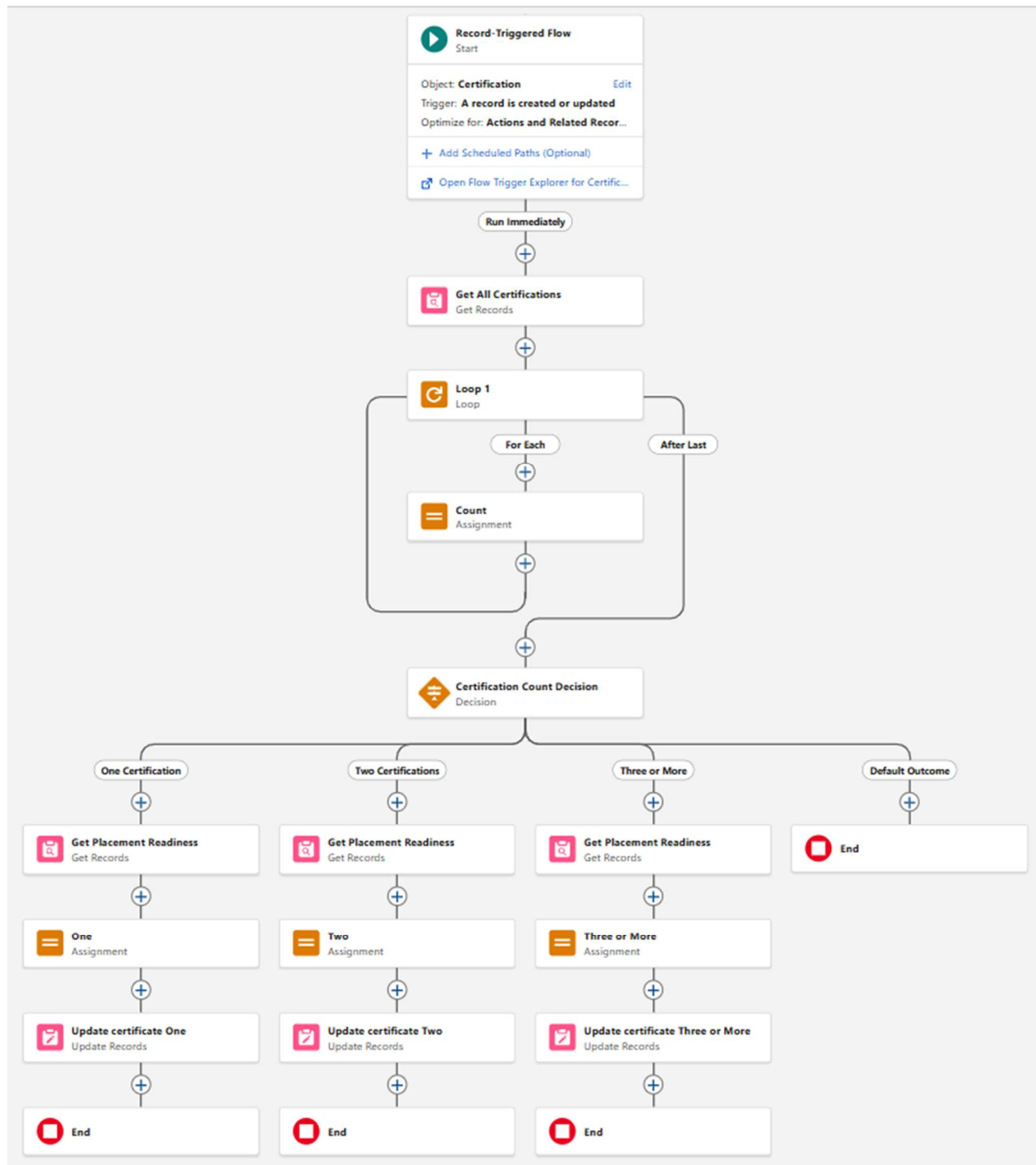


Fig: Placement Readiness Record with Updated Certification Score

CONCLUSION

The **Student Placement Readiness Management System** successfully automates the evaluation of student placement preparedness using Salesforce's powerful **declarative tools** such as custom objects, record-triggered flows, reports, and dashboards. By eliminating manual calculations and reducing human intervention, the system ensures **greater accuracy, consistency, and transparency** in placement readiness assessment.

The project provides **real-time insights** into student performance by continuously updating readiness scores whenever skills or certifications are added. This enables placement coordinators to quickly identify placement-ready candidates, monitor branch-wise readiness, and make **data-driven decisions**. The solution follows Salesforce's "**Clicks Not Code**" approach, making it easy to maintain, scalable, and adaptable to changing institutional requirements. Overall, the system demonstrates the effectiveness of Salesforce as a robust platform for **educational data management and automation**.

FUTURE SCOPE

The current system lays a strong foundation for advanced placement management. The following enhancements can be implemented in the future to further improve functionality and scalability:

- **Student Self-Service Portal**
Students can be provided with a portal to view their readiness score, track skill gaps, upload certifications, and receive personalized improvement recommendations.
- **Integration with Online Learning Platforms**
Integration with platforms such as Coursera, Udemy, or NPTEL can allow automatic fetching of certification data, reducing manual entry.
- **AI-Based Readiness Prediction (Salesforce Einstein)**
Salesforce Einstein can be used to predict placement readiness trends, recommend skills to improve, and identify students at risk of falling behind.
- **Automated Interview Scheduling**
The system can be extended to schedule interviews automatically for ready students and notify them via email or SMS.
- **Mobile Access for Placement Teams**
A mobile-friendly interface or Salesforce mobile app configuration can enable placement coordinators to track readiness and updates on the go.